MEASUREMENT INSTRUMENTS / 2026

DATUM

Measure. Compare. Specify.



Measurement instruments and selection guidance

FOUR METHODS / ONE CLEAR PURCHASING BRIEF

01 / RANGE GUIDE

Choose the right instrument.

Four measurement methods, arranged around the task rather than the product name.

Task	Instrument	Find it
Inside / outside dimensions	Digital caliper	04–05
Small outside dimensions	Outside micrometer	06–07
Relative movement	Dial indicator	08–09
Quick linear checks	Steel rule	10

ABOUT THIS PUBLICATION

Product codes and ordering information describe the catalogue specification model. Confirm final dimensions and availability with the supplier.

A measurement is a process.

Resolution, measuring range and accuracy describe different things. They should never be treated as interchangeable.

Range

The range defines the interval an instrument can measure. Select a range that comfortably covers the workpiece without making the instrument awkward to handle.

Resolution

Resolution is the smallest displayed increment or readable scale division. A finer display does not, by itself, establish better accuracy.

Accuracy

Use the manufacturer’s stated accuracy under its specified conditions. A catalog photograph cannot establish that value.

Application

Consider access, contact geometry and the surface being measured. Inside jaws, outside jaws and a depth rod serve different tasks.

Repeatability

Repeated readings help reveal changes caused by handling, alignment or the condition of the contact surfaces.

Verification

Confirm the exact model, calibration status and documentation before purchasing or using an instrument for a critical inspection.

SPECIFICATION RULE / Verify values against the actual supplier datasheet.

Digital calipers

An accessible format for outside, inside and depth measurements.



Jacek Halicki / CC BY-SA 4.0

A sliding jaw and a digital readout make the caliper easy to scan during repeated checks. Clean the contact surfaces, confirm the zero and keep the jaws square to the workpiece.

The information buyers need.

Illustrative catalog records. Confirm all purchase specifications with the supplier.

Item code	Range	Readout	Documentation
DT-C150	0–150 mm	Digital	Supplier datasheet
DT-C200	0–200 mm	Digital	Supplier datasheet
DT-C300	0–300 mm	Digital	Supplier datasheet

CHECK BEFORE ORDERING

Jaw depth, data output, battery type and environmental rating can vary between models. State those requirements in the enquiry.

SUPPLIED INFORMATION

Request the exact model reference, calibration options, warranty conditions and dimensional drawing where required.

Outside micrometers

A focused instrument for measuring small outside dimensions.



R. Henrik Nilsson / CC BY 4.0

The frame, anvil and spindle form a controlled contact arrangement. Where a ratchet or friction device is fitted, follow the manufacturer’s method for applying consistent measuring force.

Specify the contact method.

Illustrative item codes and nominal ranges; the photograph is a historical instrument example.

Item code	Nominal range	Contact form
DT-M025	0–25 mm	Flat anvil / spindle
DT-M050	25–50 mm	Flat anvil / spindle
DT-M075	50–75 mm	Flat anvil / spindle
DT-M100	75–100 mm	Flat anvil / spindle

APPLICATION NOTE

Material temperature, measuring force and cleanliness can influence a reading. Allow the part and instrument to reach suitable conditions, then follow the inspection procedure agreed for the job.

Dial indicators

Make small changes visible during alignment and comparison checks.



R. Henrik Nilsson / CC BY 4.0

Set up the contact point

The mounting arrangement should hold the instrument securely without disturbing the workpiece.

Establish a reference

An indicator compares movement against a chosen starting position. It is not a substitute for a suitable absolute dimension measurement.

Confirm the travel

Check the available movement before taking a reading.

PHOTO NOTE

Historical mechanical indicator. The illustration is not a product endorsement.

Mounting matters.

Use this checklist to collect the requirements for a quotation.

Requirement	Record before ordering
Working travel	Required movement and available clearance
Contact point	Shape, orientation and surface access
Mounting	Stem / lug format and stand compatibility
Scale or readout	Display format and readable increment
Documentation	Model reference and calibration requirement

A well-specified mounting arrangement can be as important as the indicator itself. Include a photograph or sketch of the setup when requesting a custom recommendation.

Steel rules

A simple tool that benefits from clear scale presentation.



Qurren / CC BY-SA 4.0

State length, graduation system, flexibility and edge form. A rule is useful for quick checks, while tighter inspection tolerances may require a different instrument.

07 / ENQUIRY FORM

A cleaner purchasing brief.

Complete the fields below before asking a supplier to recommend an instrument.

Field	Your requirement
Part / application	
Dimension or movement	
Required measuring range	
Acceptance tolerance	
Contact and access limits	
Calibration documentation	
Quantity / delivery date	

For final product selection, confirm the specification with the supplier.

Credits & production

CALIPER: Jacek Halicki

CC BY-SA 4.0

https://commons.wikimedia.org/wiki/File:2020_Suwmiarka_cyfrowa.jpg

https://creativecommons.org/licenses/by-sa/4.0

MICROMETER: R. Henrik Nilsson

CC BY 4.0

https://commons.wikimedia.org/wiki/File:Mid_20th_century_outside_micrometer_0_to_25_mm_0_01_mm_resolution_by_CE_Johansson_AB_Eskilstuna_Sweden.jpg

https://creativecommons.org/licenses/by/4.0

DIAL: R. Henrik Nilsson

CC BY 4.0

https://commons.wikimedia.org/wiki/File:Mid_1980s_Mahr_mechanical_dial_indicator_range_10_mm_accuracy_0_01_mm_6_rubis.jpg

https://creativecommons.org/licenses/by/4.0

RULER: Qurren

CC BY-SA 4.0

https://commons.wikimedia.org/wiki/File:Shinwa_Rules_ruler_13005.jpg

https://creativecommons.org/licenses/by-sa/4.0

Layout: Ayobami. Photography remains credited to its respective authors. Crops and JPEG conversion only. Source images retain their listed licences. No agency or manufacturer affiliation is claimed.